Kartik Khandelwal

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leetcode.com/u/Kartik_Khandelwal_2004 | my-portfolio-azure-nine-26.vercel.app

Education

KIET Group of Institutions, Ghaziabad 2024 – 2028 (Expected)
B.Tech in Information Technology | CGPA: 8.17 | Coursework: DSA, OOP, Operating Systems, DBMS

Summary

GenAI engineer in training who builds LLM apps end to end and measures them: a RAG system over 68 hours of lectures (hit@5 65% → 95%), a resume-grounded recruiter chatbot with 31 tests, and an Android app with an AI assistant. Python, FastAPI, Groq, Qdrant; 4 AWS certifications; 450+ LeetCode problems. Seeking a GenAI / Applied AI Engineer internship.

Technical Skills

Languages: Python, C++, Kotlin, TypeScript/JavaScript, Java
GenAI & LLMs: RAG, Prompt Engineering, Embeddings, Vector Search (Qdrant), Query Rewriting, Structured Outputs (JSON Schema, Pydantic), Streaming, LLM Evaluation (hit@k, MRR, golden sets), Guardrails & Prompt-Injection Defense, Caching, Groq & Gemini APIs
Backend & Web: FastAPI, REST APIs, React, Vite, Tailwind; **Testing:** pytest, Vitest, Playwright
Cloud & DevOps: AWS (Lambda, API Gateway, SQS, DynamoDB, EC2, VPC), Terraform, GitHub Actions, Cloudflare Workers, Vercel, Render, Supabase, Firebase, uv, Git
Android: Jetpack Compose, MVVM, Clean Architecture, Hilt, Room, Retrofit, Coroutines, Flow, Play Billing

Projects

YT Lecture RAG — Timestamp-Level Search over DSA Lectures Live Demo → | Code →
Python · FastAPI · Qdrant · fastembed (ONNX) · Groq · Vercel

- Built search over 126 YouTube DSA lectures (∼68 hrs, 2,933 chunks) that jumps to the exact second a topic is explained; transcripts are chunked into 75 s windows with 15 s overlap.
- Diagnosed that the English-only embedder matched Hinglish queries by language, not topic; an LLM query-rewrite step raised hit@5 from 65% to 95% and MRR from 0.59 to 0.77 on a 20-question golden set.
- Grounded answers with three refusal guards (distance cutoff, model refusal, citation check), refusing 8/8 off-topic questions; moved embeddings to ONNX to fit Vercel serverless with no metric change.

HireMeAI — AI Recruiter Chatbot Live Demo → | Code →
Python · FastAPI · Groq (gpt-oss-120b) · Pydantic · React · pytest · Render

- Built a chatbot that answers recruiter questions only from my resume, with streamed responses; the resume is parsed once into a strict JSON schema and cached by file, prompt and model, so startup makes zero LLM calls.
- Wrote a golden-set eval that caught silent data loss from schema gaps and a brittle post-processing filter; 31 pytest tests with the LLM mocked run in under 2 seconds.
- Hardened it for public use: phone number withheld from the model, prompt-injection and XSS defenses, input caps, per-IP and global rate limits; found an X-Forwarded-For spoofing gap and bounded it with the global cap.

MealTime — AI Recipe & Meal-Planning Android App Code →
Kotlin · Jetpack Compose · Hilt · Room · Firebase · Gemini/Groq · Cloudflare Workers · Play Billing

- Built recipe search, cooking mode, favorites and a smart shopping list with MVVM + Clean Architecture, a 5-table Room cache and a bundled offline fallback.
- Added AI recipe generation, recipe remix and a multi-day meal planner, routed through a Cloudflare Worker authenticated with Firebase ID tokens; Gemini with automatic Groq fallback, and no API keys in the APK.
- Enforced subscriptions server-side by validating Play purchase tokens and setting Firebase custom claims; GitHub Actions runs build, lint and unit tests on every push.

DSA Practice Tracker — Spaced-Repetition PWA Live Demo → | Code →
React · Vite · Tailwind · Supabase · Vitest · Playwright · Vercel

- Built an installable, offline-first tracker for 917 DSA problems across 23 topics with 7/30/90-day spaced review; a failed review drops back to a 3-day re-check.
- Added optional cross-device sync on Supabase (Google sign-in, row-level security, conflict merge), covered by Vitest unit and Playwright end-to-end tests.

More Android apps: Kometo (food delivery; Hilt, Firebase Auth, 8+ Compose screens), MyWishlist (offline; Room, Flow).

Achievements & Certifications

- Solved **450+ DSA problems** on LeetCode, with a 100+ day coding streak.
- National-level participant, Bharatiya Antariksh Hackathon 2025 (ISRO & Hack2Skill).
- **AWS Certified CloudOps Engineer – Associate** and **Data Engineer – Associate** (Apr 2026); AWS Certified AI Practitioner and Cloud Practitioner (Dec 2025).